Network Security and Monitoring Report

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Date : 22 February 2026

Target Machine : Linux (VM)

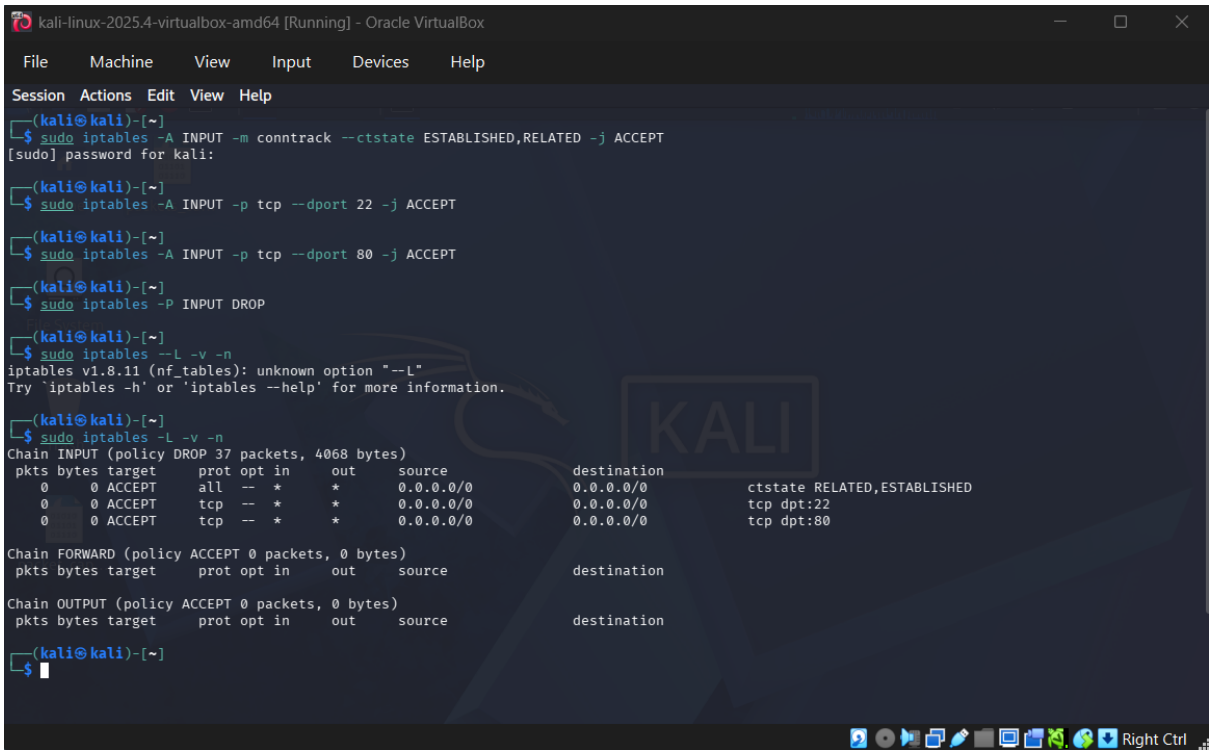
Introduction:

- The objective of this task was to gain practical experience in network monitoring and defence by implementing a firewall, capturing network traffic, and performing a system vulnerability audit.
- This project focuses on gaining hands-on experience in essential cybersecurity domains, specifically network traffic monitoring, defensive system configuration, and proactive security auditing.
- The objective is to demonstrate how network monitoring using Scapy and firewall management via iptables work together to secure a Linux environment against unauthorized access.
- A key component of the task involves identifying and documenting system weaknesses using automated scanning tools to understand the severity and potential impact of security vulnerabilities on a network.

Firewall Configuration (Using iptables) :

- **Tool Used :** iptables.
- **Methodology :** I configured the Linux firewall to allow traffic only on specific ports required for services.

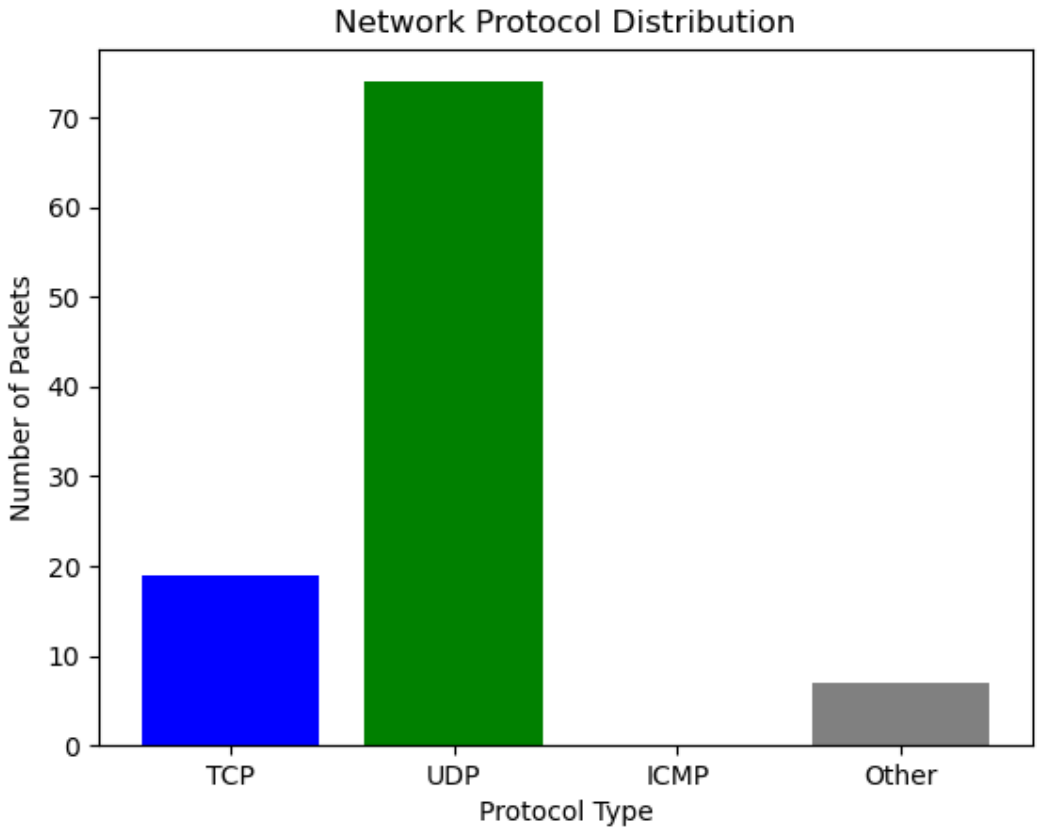
- **Rules Implemented :**
    1. Allowed SSH on port 22.
    2. Allowed HTTP on port 80.
    3. Set the default policy to DROP for all other incoming traffic.



Packet Sniffing and Analysis (Using Scapy) :

- **Tool Used :** Python, Scapy library, Matplotlib.
- **Methodology :** A Python script was developed to capture 100 packets from the local network interface.
- **Findings :**

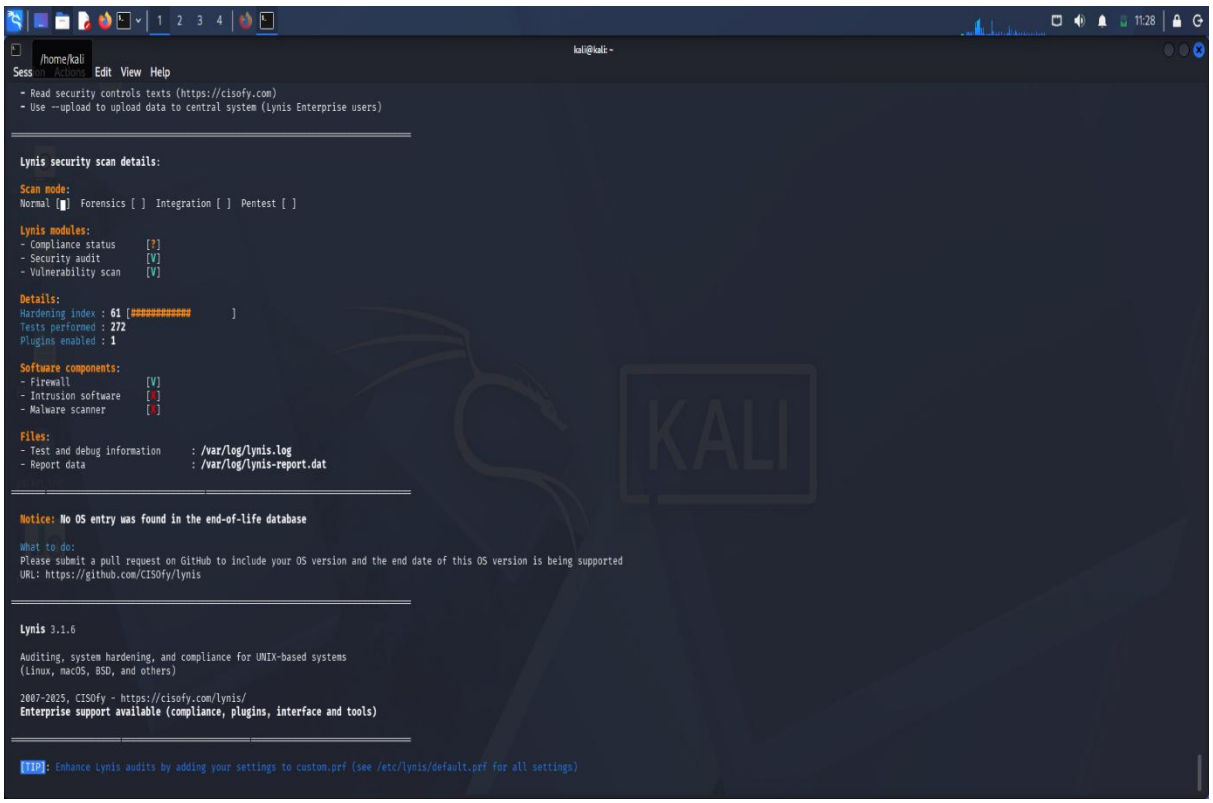
1. **UDP:** The most frequent protocol, accounting for over 70% of traffic.
2. **TCP:** Identified as standard web/connection traffic.
3. **ARP/Other:** Broadcast traffic was captured as evidenced in the Wireshark log.



Vulnerability Scanning (Using Lynis) :

- **Tool Used :** Lynis (System Auditing Tool).
- **Methodology :** Performed a comprehensive system audit to identify security weaknesses and hardening opportunities.

- **Results :**
  1. **Hardening Index:** My system received a score **61.**
  2. **Suggestions:**
     - The scan suggested improving password aging policies and clearing old software packages.
     - You should disable **"Root Login"** and **"X11 Forwarding"** if they aren't needed.



Key Learnings :

- I learned how iptables rules act as a first line of defence.
- how scapy provides deep visibility into packet headers.
- Learned to implement sudo codes efficiently.

- Learned to update packages regularly.
- Learned how the system works correctly.

Key Challenges :

- Setting up the vulnerability scanner (OpenVAS) was time-consuming due to feed syncs and complexify setup.
- After staying committed by putting continuous efforts 4 days, I gave up with OpenVAS because of deadline & pending tasks. So, I used lynis as an alternative.